1. ICs Classification

Integrated circuits offer a wide range of applications and could be broadly classified as:

Digital ICs

Linear ICs

Based upon the above requirements, two distinctly different IC technology namely, Monolithic technology and Hybrid technology have been developed. In monolithic integrated circuits, all circuit components, both active and passive elements and their interconnections are manufactured into or on top of a single chip of silicon. The monolithic circuit is ideal for applications where identical circuits are required in very large quantities and hence provides lowest per-unit cost and highest order of reliability. In hybrid circuits, separate component parts are attached to a ceramic substrate and interconnected by means of either metallization pattern or wire bonds.

A monolithic integrated circuit (IC), commonly referred to as a "monolithic IC" or simply "IC," is a type of electronic circuit that is constructed on a single semiconductor substrate or chip. This chip typically consists of various electronic components, such as transistors, resistors, capacitors, and diodes, all fabricated on the same piece of semiconductor material, typically silicon.

Here are some key characteristics and advantages of monolithic integrated circuits:

1. **Compactness**: Monolithic ICs are incredibly compact because all the components are integrated onto a single chip. This miniaturization allows for the creation of complex electronic functions within a very small physical space.

2. **Reliability**: Since all components are fabricated together on the same substrate, there are fewer interconnections, reducing the risk of electrical and mechanical failures that can occur in discrete circuits with numerous external connections.

3. **Cost-Efficiency**: Mass production of monolithic ICs can lead to cost savings, as the same manufacturing processes can be used to produce a large number of identical chips.

4. **Power Efficiency**: Monolithic ICs are often designed to be power-efficient, making them suitable for battery-operated devices and other applications where power consumption is a concern.

5. **High Performance**: These circuits can provide high-speed performance, making them suitable for applications like microprocessors, memory chips, and other complex electronic systems.

6. **Customization**: Manufacturers can design monolithic ICs for specific functions or applications by customizing the layout and components on the chip.

7. **Reduced Size and Weight**: Compared to discrete components, monolithic ICs can significantly reduce the size and weight of electronic devices.

Monolithic ICs have revolutionized the electronics industry, enabling the development of a wide range of electronic devices, from simple analog amplifiers to complex digital processors. They are a fundamental building block in modern electronics, and their continued miniaturization and innovation have contributed to the rapid advancement of technology

Hybrid integrated circuits, often referred to as "hybrid ICs," are a type of electronic circuit in which separate component parts, such as transistors, resistors, capacitors, and diodes, are attached to a substrate, typically made of ceramic or some other insulating material. These individual components are then interconnected using various techniques, including metallization patterns or wire bonds.

The term "hybrid" in hybrid integrated circuits refers to the combination of discrete or separate components with integrated elements on a single substrate. This approach allows for the benefits of both integrated circuits (ICs) and discrete components to be leveraged in a single package.

Key characteristics and advantages of hybrid integrated circuits include:

1. **Customization**: Hybrid ICs are often used for custom applications where specific components and functions are required. Manufacturers can select and arrange discrete components to meet the unique needs of a particular circuit.

2. **Mixed Technology**: They combine the advantages of integrated circuits (compactness, reliability, and high performance) with discrete components' flexibility and ease of customization.

3. **Small Quantity Production**: Hybrid ICs are suitable for small-scale or low-volume production because they allow for flexibility in component selection and arrangement.

4. **Reliability**: The use of ceramic substrates and well-controlled manufacturing processes can lead to high reliability and long-term stability.

5. **Wide Range of Applications**: Hybrid ICs can be found in various applications, including aerospace, telecommunications, medical devices, and specialized industrial equipment.

While hybrid integrated circuits offer flexibility and customization benefits, they may not be as cost-effective or as compact as monolithic integrated circuits (ICs) for high-volume production of standardized electronic devices. Nonetheless, they are essential in situations where specific requirements or small-scale production are necessary.

here's a comparison table between monolithic technology and hybrid technology:

Aspect	Monolithic Technology	Hybrid Technology
Definition	Entire circuit fabricated on a single substrate.	Combination of multiple technologies/components.
Integration Level	High integration level, all components on a chip.	Moderate integration, combines discrete parts.
Fabrication Cost	Typically lower cost due to batch production.	Often higher cost due to multiple components.
Size and Weight	Smaller size and lighter weight.	Larger size and heavier due to discrete parts.
Performance	Better performance due to shorter interconnections.	May have performance limitations.
Power Efficiency	Usually more power-efficient.	May have higher power consumption.
Customization	Less customizable, fixed design on a chip.	More customizable, mix and match components.
Reliability	Generally more reliable due to fewer connections.	May be less reliable due to more connections.

Repairability	Difficult to repair or modify once fabricated.	Easier to repair or modify individual components.
Examples	Microprocessors, ICs, CMOS technology.	Space missions, RF circuits, power amplifiers.

Keep in mind that the choice between monolithic and hybrid technology depends on the specific application and requirements, as each has its own advantages and disadvantages.

IC CHIP SIZE AND CIRCUIT COMPLEXITY

UP until the 1950s, the electronic device technology was dominated by the vacuum tube. The present day electronics is the result of the invention of the transistor in 1947. The invention of the transistor by William B. Shockley, Walter H. brattain and John Bardeen of Bell Telephone Laboratories was followed by the development of the Inte- grated circuit (IC). The concept of IC was introduced at the beginning of 1960 by both Texas Instruments and Fairchild Semiconductors. Since that time, the size and complexity of ICs have increased rapidly as shown by the brief chronology.

1.4 FUNDAMENTALS OF MONOLITHIC IC TECHNOLOGY

A monolithic circuit, literally speaking, means a circuit fabricated from a single stone or a single crystal. The origin of the word 'mono- lithic' is from the Greek word monos meaning 'single' and lithos meaning 'stone'. So monolithic integrated circuits are, in fact, made in a single piece of single crystal silicon. The most significant advantage of integrated circuit of reducing the cost of production of electronic circuits due to batch production can be easily visualized by a simple example. A standard 10 cm diameter wafer can be divided into approximately 8000 rectangular chips of sides 1 mm. Each IC chip may contain as few as tens of components to several thousand components. And if 10 such wafers are processed in one batch, we can make 80,000 ICs simultaneously. Many chips so produced will be faulty due to imperfection in the manufacturing process. Even if the yield (percentage of fault free chips/wafer) is only 20 percent, it can be seen that 16,000 good chips are produced in a single batch. The fabrication of discrete devices such as transistor, diode or an integrated circuit in general can be done by the same technology. The various processes usually take place through a single plane and there- fore, the technology is referred to as planar technology. A simple circuit of Fig. 1.3 when fabricated by silicon planar technology will have the cross-sectional view shown in Fig. 1.4.

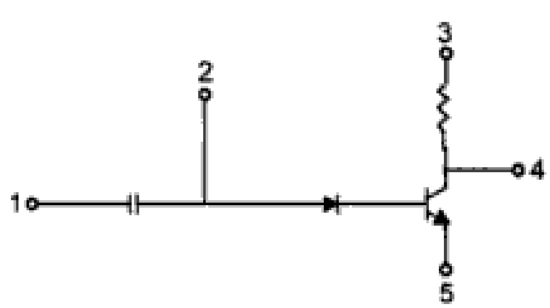


Fig. 1.3

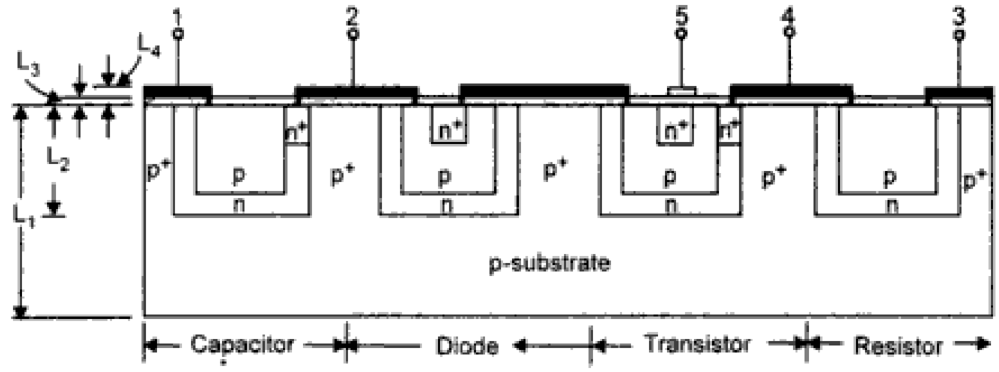


Fig. 1.4 Complete cross-sectional view of the circuit in Fig. 1.3 when transformed into monolithic form

An IC in general, consists of four distinct layers, as follows:

Layer No. 1 (-400 µm)

is a p-type silicon substrate upon which the integrated circuit is fabricated.

Layer No. 2 (~5-25 µm)

is a thin n-type material grown as a single crystal extension of the substrate using epitaxial deposition technique. All active and passive components are fabricated within this layer using selective diffusion of impurities.

Layer No. 3 (0.02-2µm)

is a very thin SiO2 layer for preventing diffusion of impurities wherever not required using photo lithographic technique.

Layer No. 4 (-lum) is an aluminium-layer used for obtaining interconnection between components.

It may be pointed out that the drawings showing the cross-sec- tional view in this chapter are never scale drawings, but are distorted for the particular emphasis required.

1.5 BASIC PLANAR PROCESSES

The basic processes used to fabricate ICs using silicon planar technology can be categorised as follows:

1. Silicon wafer (substrate) preparation
2. Epitaxial growth
3. Oxidation
4. Photolithography
5. Diffusion
6. Ion implantation
7. Isolation technique
8. Metallization
9. Assembly processing and packaging

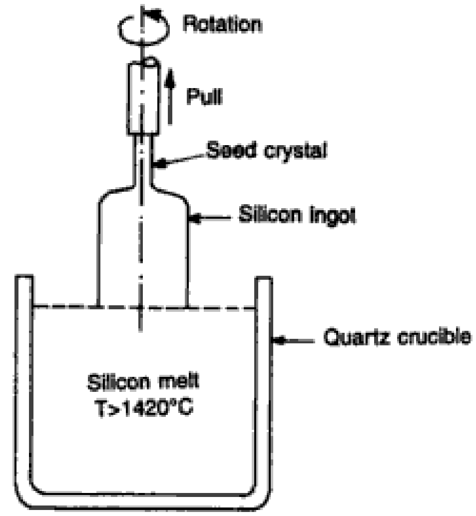
We shall now describe these processes in detail

1.5.1 Silicon Wafer Preparation:

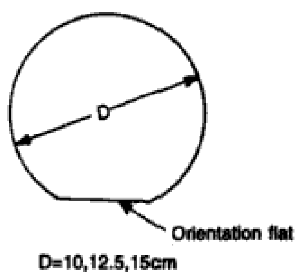
The following steps are used in the preparation of Si-wafers

1. Crystal growth and doping
2. Ingot trimming and grinding
3. Ingot slicing
4. Wafer polishing and etching
5. Wafer cleaning

The starting material for crystal growth is highly purified (99.99999) polycrystalline silicon. The Czochralski crystal growth process is the most often used for producing single crystal silicon ingots. The polycrystalline silicon together with an appropriate amount of dopant is put in a quartz crucible and is then placed in a furnace. The material is then heated to a temperature in excess of the silicon melting point of 1420°C. A small single crystal rod of silicon called a seed crystal is then dipped into the silicon-melt and slowly pulled out as shown in Fig. 1.5. As the seed crystal is pulled out of the melt, it brings with it a solidified mass of silicon with the same crystalline structure as that of seed crystal. During the crystal pulling process, the seed crystal and the crucible are rotated in opposite directions in order to produce ingots of circular cross-section. The diameter of the ingot is controlled by the pulling rate and the melt temperature. Ingot diameter of about 10 to 15 cm is common and ingot length is generally of the order of 100 cm.



Next the top and bottom portions of the ingot are cut off and the ingot surface is ground to produce an exact diameter (D= 10, 12.5, 15 cm). The ingot is also ground flat slightly along the length to get a reference plane. The ingot is then sliced using a stainlesssteel saw blade with industrial diamonds embedded into the inner diameter cutting edge. This produces circular wafers or slices as shown in Fig. 1.6. The orientation flat portion serves as a useful reference plane for the various processes described later. The silicon wafers so obtained have very rough surface due to slicing operation. These wafers undergo a number of polishing steps to produce a flat surface. Then one side of the wafer is given a final mirror-smooth highly polished finish, whereas the other side is simply lapped on an abrasive lapping machine to obtain an acceptable degree of flatness. Finally, the wafers are thoroughly rinsed and dried. A raw cut slice of thickness 23-40 mils produces wafers of 16-32 mils thickness after all the polishing steps.



These silicon wafers will contain several hundred rectangular chips, each one containing a complete integrated circuit. After all the IC fabrication processes are complete, these wafers are sawed into 100 to 8000 rectangular chips having side of 10 to 1 mm. Each chip is a single IC and may contain hundreds of components. The wafer thickness therefore is so chosen that it is possible to separate chips without breaking and at the same time, it gives sufficient mechanical strength to the IC chip.

1.5.2 Epitaxial Growth

The word epitaxy is derived from Greek word epi meaning 'upon' and the past tense of the word teinon meaning 'arranged'. So, one could describe epitaxy as, arranging atoms in single crystal fashion upon a single crystal substrate, so that the resulting layer is an extension of the substrate crystal structure.

The basic chemical reaction used for the epitaxial growth of pure silicon is the hydrogen reduction of silicon tetrachloride.

$SiC + 2H_2$ 1200°C. $Si + 4HCI$

Mostly, epitaxial films with specific impurity concentration are required. This is accomplished by introducing phosphine (PH,) for the n-type and bi-borane (BH) for p-type doping into the silicon-tetrachloride hydrogen gas stream.

The process is carried out in a reaction chamber consisting of a long cylindrical quartz tube encircled by an RF induction coil. Figure 1.7 shows the diagrammatic representation of the system used. The silicon wafers are placed on a rectangular graphite rod called a boat. This boat is then placed in the reaction chamber where the graphite is heated inductively to a temperature 1200°C. The various gases required for the growth of desired epitaxial layers are introduced into the system through a control console

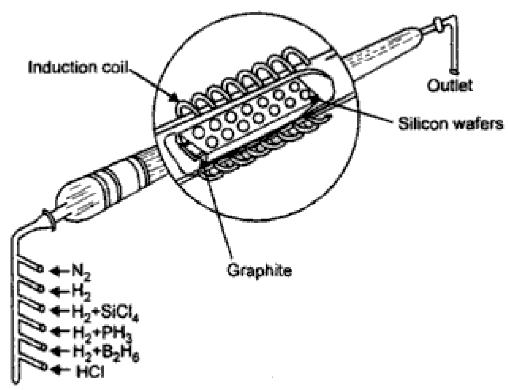


Fig. 1.7 a diagrammatic representation of a system for growing silicon epitaxial films.

1.5.3 Oxidation

SiO, has the property of preventing the diffusion of almost all impu- rities through it. It serves two very important purposes.

1. SiO_2 is an extremely hard protective coating and is unaffected by almost all reagents except hydrofluoric acid. Thus it stands against

any contamination. 2. By selective etching of SiO,, diffusion of impurities through care- fully defined windows in the SiO, can be accomplished to fabri- cate various components. The silicon wafers are stacked up in a quartz boat and then in-serted into quartz furnace tube. The Si-wafers are raised to a high temperature in the range of 950 to 1115°C and at the same time, exposed to a gas containing O, or H_2O or both. The chemical reaction is:

$$Si + 2H_2O \rightarrow\rightarrow\rightarrow SiO_2 + 2H_2$$

This oxidation process is called thermal oxidation because high temperature is used to grow the oxide layer. The thickness of the film is governed by time, temperature and the moisture content. The thick- ness of oxide layer is usually in the order of 0.02 to 2 μm.

1.5.4 Photolithography

With the help of photolithography, it has become possible to produce microscopically small circuit and device patterns on Si-wafers. As many as 10,000 transistors can be fabricated on a 1 cm x 1 cm chip. The conventional photolithographic process uses ultraviolet light exposure and device dimension or line width as small as 2 um can be obtained. However, with the advent of latest technology using X-ray or electron beam lithographic techniques, it has become possible to produce de- vice dimension down to submicron range (< 1 μm).

Photolithography involves two processes, namely: Making of a photographic mask

Photo etching

The making of a photographic mask involves the following sequence of operations-first the preparation of initial artwork and secondly. its reduction. The initial layout or artwork of an IC is normally done at a scale several hundred times larger than the final dimensions of the finished monolithic circuit. This is because, for a tiny chip, larger the artwork, more accurate is the final mask. For example, it is often required to make an opening of width about 1 mil (25 μm). Obviously, this cannot be managed by any draftsman even with his thinnest of sketch pens. So the drawings are made magnified and often by a factor of 500. With this magnification, it is easy to see that a width of one mil is magnified to a width of 500 mils, that is, about 1.2 cm. Therefore, for a finished monolithic chip of area 50 mil x 50 mil, the artwork will be made on an area of about 60 cm x 60 cm.

This initial layout is then decomposed into several mask layers, each corresponding to a process step in the fabrication schedule, e.g.. a mask for base diffusion, another for collector diffusion, another for metallization and so on.

For photographic purpose, artwork should not contain any line drawings but must be of alternate clear and opaque regions. This is accomplished by the use of clear Mylar coated with a sheet of red photographically opaque mylar (trade name-Rubylith). The red layer can be easily peeled off thus exposing clear areas with a knife edge from the regions where impurities have to be diffused. The artwork is usually produced on a precision drafting machine, known as coordinatograph. The coordinatograph has a cutting head that can be positioned accurately and moved along two perpendicular axes. The coordinatograph outlines the pattern cutting through the red mylar without damaging the clear layer underneath.

This rubylith pattern of individual mask is photographed and then reduced in steps by a factor of 5 or 10 several times to finally obtain the exact image size. The final image also must be repeated many times in a matrix array, so that many ICs will be produced in one process. The photo repeating is done with a step and repeat camera. This is an imaging device with a photographic plate on a movable platform. Between exposure, the plate is moved in equal steps so that successive images form in an array. When the exposed plate is devel- oped, it becomes a master maak. The masks, actually used in IC processing are made by contact printing from the master. These working masks wear out with use and are replaced as required.

Photo-etching is used for the removal of SiO, from desired regions so that the desired impurities can be diffused. The wafer is coated with a film of photosensitive emulsion (Kodak Photoresist KPR). The thickness of the film is in the range of 5000-10000 A as shown in Fig. 1.8 (a). The mask negative of the desired pattern) as prepared by steps described earlier is placed over the photoresist coated wafer as shown in Fig. 1.8 (b). This is now exposed to ultraviolet light, so that KPR becomes polymerized beneath the transparent regions of the mask. The mask is then removed and the wafer is developed using a chemical (trichloroethylene) which dissolves the unexposed/ unpolymerized regions on the photoresist and leaves the pattern as shown in Fig. 1.8 (c). The polymerised photoresist is next fixed or cured, so that it becomes immune to certain chemicals called etchants used in subsequent processing steps. The chip is immersed in the etching solution of hydrofluoric acid, which removes the SiO, from the areas which are not protected by KPR as shown in Fig. 1.8 (d) After diffusion of impurities, the photoresist is removed with a chemical solvent (hot H,SO,) and mechanical abrasion.

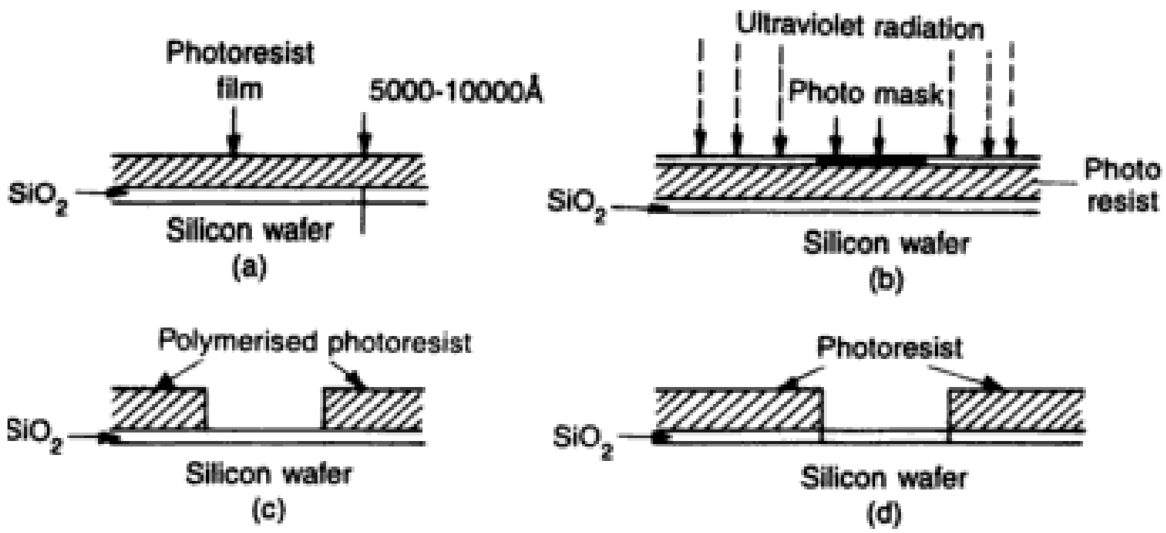


Fig. 1.8 various step for light etching.

The etching process described is a wet etching process and the chemical reagents used are in liquid form. A new process used these days is a dry etching process called plasma etching. A major advantage of the dry etching process is that it is possible to achieve smaller line openings (≤ 1 um) compared to wet process. Complete description of the plasma etching process is beyond the scope of this book.

X-Ray and Electron Beam Lithography

With conventional ultraviolet (UV) photolithography process in which the UV wavelengths used are in the range 0.3 to 0.4 μm, the mini- mum device dimensions or line widths are limited by diffraction effects to around five wavelengths or about 2 um. This is what puts an upper limit on the IC device density using UV photolithography With the advent of X-ray and electron beam lithography techniques, it has become possible to produce device dimensions down to submicron range (< 1 μm). This is due to much shorter wavelengths involved. With these techniques, MOSFET with gate length as small as 0.25 um have been made. The cost of X-ray or electron beam equipment is very high and the exposure times very much longer than with UV photolithography. So this becomes a very expensive process and is used only when very small device dimensions (51 μm) are needed.

1.5.5 Diffusion

Another important process in the fabrication of monolithic ICs is the diffusion of impurities in the Silicon chip. This uses a high temperature furnace having a flat temperature profile over a useful length (about 20" length). A quartz boat containing about 20 cleaned wafers is pushed into the hot zone with temperature maintained at about a 1000°C. Impurities to be diffused are rarely used in their elemental forms. Normally, compounds such as B_2O, (Boron oxide), BCI, (Boron chloride) are used for Boron and P,O, (Phosphorous pentaoxide) and POCI, (Phosphorous oxychloride) are used as sources of Phosphorous. A carrier gas, such as dry oxygen or nitrogen is normally used for sweeping the impurity to the high temperature zone. The depth of diffusion depends upon the time of diffusion which normally extends to 2 hours.

The diffusion of impurities normally takes place both laterally as well as vertically. Therefore, the actual junction profiles will be curved as shown in Fig. 1.9. However, for the sake of simplicity, lateral diffusion will be omitted in all the drawings.

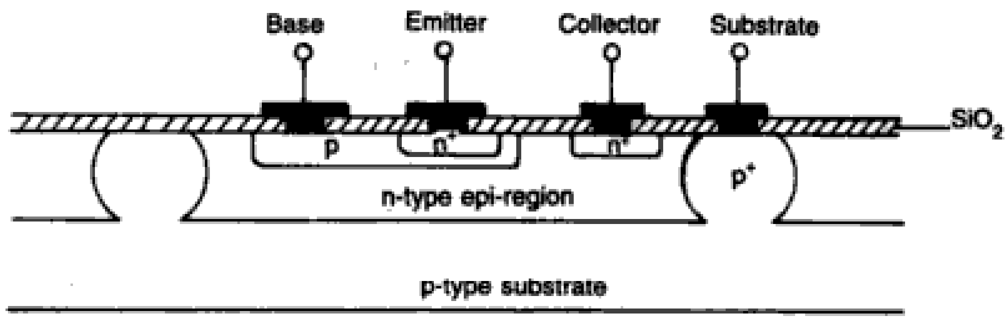


Fig. 1.9 The cross-section of an npn transistor showing curved junction profiles as a result of lateral diffusion.

1.5.6 Ion Implatation

Ion implantation is the other technique used to introduce impurities into a silicon wafer. In this process, silicon wafers are placed in a vacuum chamber and are scanned by a beam of high-energy dopant ions (borons for p-type and phosphorus for n-type) as shown in Fig. 1.10. These ions are accelerated by energies between 20 kV to 250 kV. As the ions strike the silicon wafers, they penetrate some small dis- tance into the wafer. The depth of penetration of any particular type of ion increases with increasing accelerating voltage. Ion implantation technique has two important advantages.

1. It is performed at low temperatures. Therefore, previously dif fused regions have a lesser tendency for lateral spreading.

2. In diffusion process, temperature has to be controlled over a large area inside the oven, whereas in ion implantation tech- nique, accelerating potential and the beam current are electri- cally controlled from outside

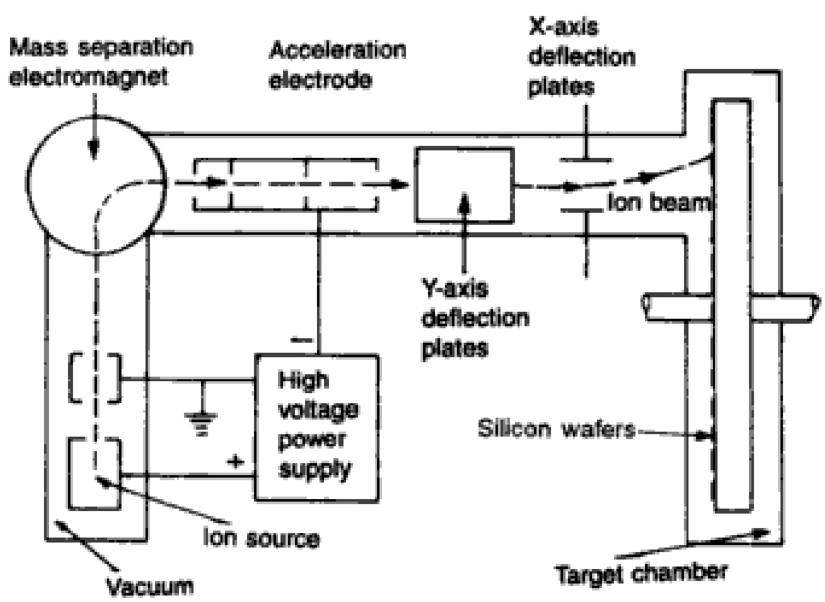


Fig. 1.10 Ion implantation system

1.5.7 Isolation Techniques

Since a number of components are fabricated on the same IC chip, it becomes necessary to provide electrical isolation between different components and interconnections. Various types of isolation techniques have been developed. However, we shall discuss here only two commonly used techniques namely:

pn junction isolation Dielectric isolation

p-n Junction Isolation

In this isolation technique, p' type impurities are selectively diffused into the n-type epitaxial layer so as to reach p-type substrate as shown in Fig. 1.11 (a). This produces islands surrounded by p-type moats. It can be seen that these regions are separated by two back- to-back p-n junction diodes. If the p-type substrate material is held at the most negative potential in the circuit, the diodes will be reverse biased providing electric isolation between these islands. The different components are fabricated in these isolation islands. The concentration of the acceptor atoms in the region between isolation islands is usually kept much higher (p) than the p-type substrate. This prevents the depletion region of the reverse biased diode from penetrating more into p' region and possibly connecting the isolation islands

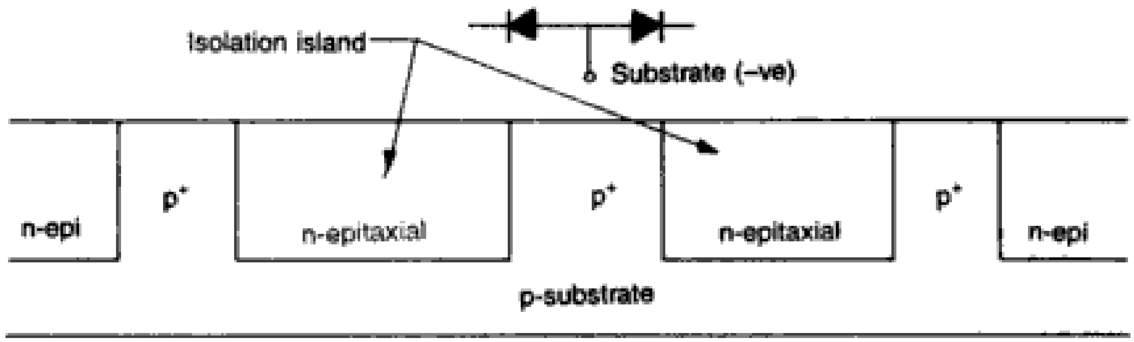


Fig 1.11 (a) p-n junction isolation

There is, however, one undesirable by-product of this isolation process. It is the presence of a transition capacitance at the isolating pn junctions, resulting in an inevitable capacitor coupling between the components and the substrate. These parasitic capacitances limit the performance of the, circuit at high frequencies. But being economical, this technique is commonly used for general purpose ICs.

Dielectric Isolation

Here a layer of solid dielectric such as silicon dioxide or ruby completely surrounds each component, thereby producing isolation, both electrical and physical. This isolating dielectric layer is thick enough so that its associated capacitance is negligible. Also, it is possible to fabricate both pnp and npn transistor within the same silicon substrate. Since this method requires additional fabrication steps, it becomes more expensive. The technique is mostly used for fabricating professional grade ICs required for specialised applications viz, aerospace and military, where higher cost is justified by superior performance. Figure 1.11 (b) shows such a structure.

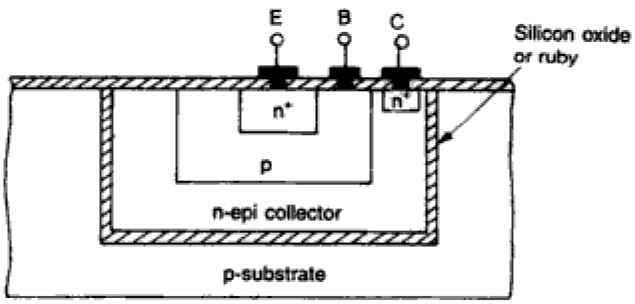


Fig. 1.11 (b) dielectric isolation

1.5.8 Metallization

The purpose of this process is to produce a thin metal film layer that will serve to make interconnections of the various components on the chip. Aluminium is usually used for the metallization of most ICs as it offers several advantages.

1. It is relatively a good conductor.

2. It is easy to deposit aluminium films using vacuum deposition.

3. Aluminium makes good mechanical bonds with silicon. 4. Aluminium forms low resistance, non-rectifying (ie. ohmic) contact with p-type silicon and the heavily doped n-type silicon.

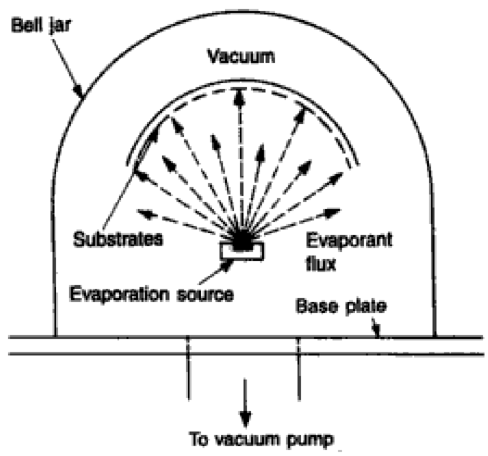


Fig. 1.12 vacuum evaporation for metallization

The film thickness of about 1 um and conduction width of about 2 to 25 um are commonly used. The process takes place in a vacuum evaporation chamber as shown in Fig. 1.12. The pressure in the chamber is reduced to the range of about 10 to 107 torr (1 atmosphere =760 torr = 760 mm Hg). The material to be evaporated is placed in a resistance heated tungsten coil or basket. A very high power density electron beam is focussed at the surface of the material to be evaporated. This heats up the material to very high temperature and it starts vaporizing. These vapors travel in straight line paths. The evaporated molecules hit the substrate and condense there to form a thin film coating.

After the thin film metallization is done, the film is patterned to produce the required interconnections and bonding pad configuration. This is done by photolithographic process and aluminium is etched away from unwanted places by using etchants like phosphoric acid (H,PO,)

1.5.9 Assembly Processing and Packaging

Each of the wafer processed contains several hundred chips, each being a complete circuit. So these chips must be separated and individually packaged. A common method called scribing and cleaving used for separation makes use of a diamond tipped tool to cut lines into the surface of the wafer along the rectangular grid separating the individual chips. Then the wafer is fractured along the scribe lines and the individual chips are physically separated. Each chip is then mounted on a ceramic wafer and attached to a suitable package. There are three different package configurations available.

1. TO-5 glass metal package

2. Ceramic flat package

3. Dual-in-line (ceramic or plastic type)

TO-5 packages are available in 8, 10 or 12 leads, whereas the flat or dual-in-line package is commonly available in 8, 14 or 16 leads, but even 24 or 36 or 42 leads are also available for special circuits.

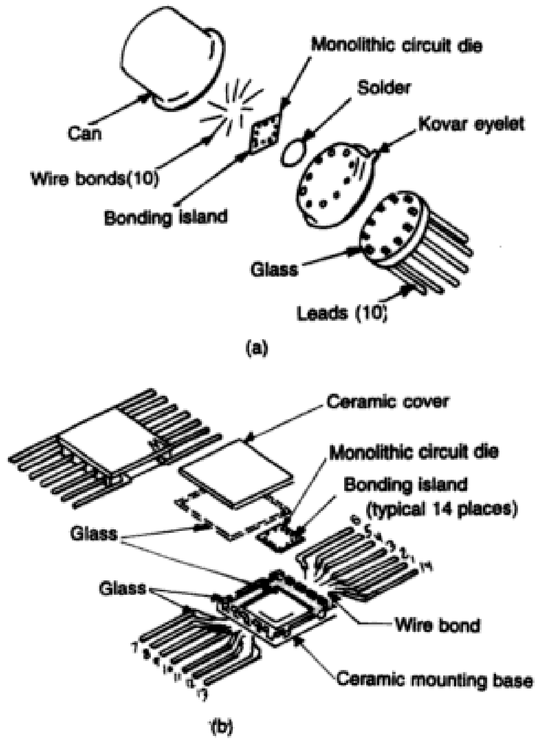


Fig. 1.13 Exploded view of (a) lead TO-5 package (b) 14-lead version of the flat package, showing the various components as well as the completed fat package

Ceramic packages, whether of flat type or dual-in-line are costly due to fabrication process, but have the advantage of best hermetic sealing. Most of the general purpose ICs are dual-in-line plastic packages due to economy. Figure 1.13 (a, b) shows the exploded view of TO-5 and flat package.

1.6 FABRICATION OF A TYPICAL CIRCUIT

We shall here show the various steps utilized in converting the circuit of Fig. 1.3 into the monolithic IC of Fig. 1.4.

Step-1: Wafer Preparation

Refer Fig. 1.14 (a). The starting material called the substrate is a p-type silicon wafer prepared as discussed in Sec. 1.5.1. The wafers are usually of 10 cm diameter and 0.4 mm(-400 µm) thickness. The resistivity is approximately 10 02-cm corresponding to concentration of acceptor atom, N, 1.4 x 1015 atoms/cm³.

Step-2: Epitaxial Growth

An n-type epitaxial film (5-25 μm) is grown on the p-type substrate as shown in Fig. 1.14 (b). The ultimately becomes the collector region of the transistor, or an element of the diode and diffused capacitor associated with the circuit. So in general it can be said that all active and passive components are fabricated within this layer. The resistivity of n-epitaxial layer is of the order of 0.1 to 0.5 0-cm.

Step 3: Oxidation

Refer Fig. 1.14 (e) A SiO, layer of thickness of the order of 0.02 to 2 μm is grown on the n-epitaxial layer.

Step 4: Isolation Diffusion

In the circuit of Fig. 1.3 four components have to be fabricated, so we require four islands which are isolated. For this, SiO, is removed from five different places using photolithographic technique. Refer Fig. 1.14 (d). The wafer is next subjected to heavy p-type diffusion for a long time interval so that p-type impurities penetrate the n-type epitaxial layer and reach the p-type substrate. The area under the SiO, are n-type islands that are completely surrounded by p-type moats. As long as the pn junctions between the isolation islands are held at reverse bias, that is, the p-type substrate is held at a negative potential with respect to the n-type isolation islands, these regions are electrically isolated from each other by two back-to-back diodes, providing the desired isolation.

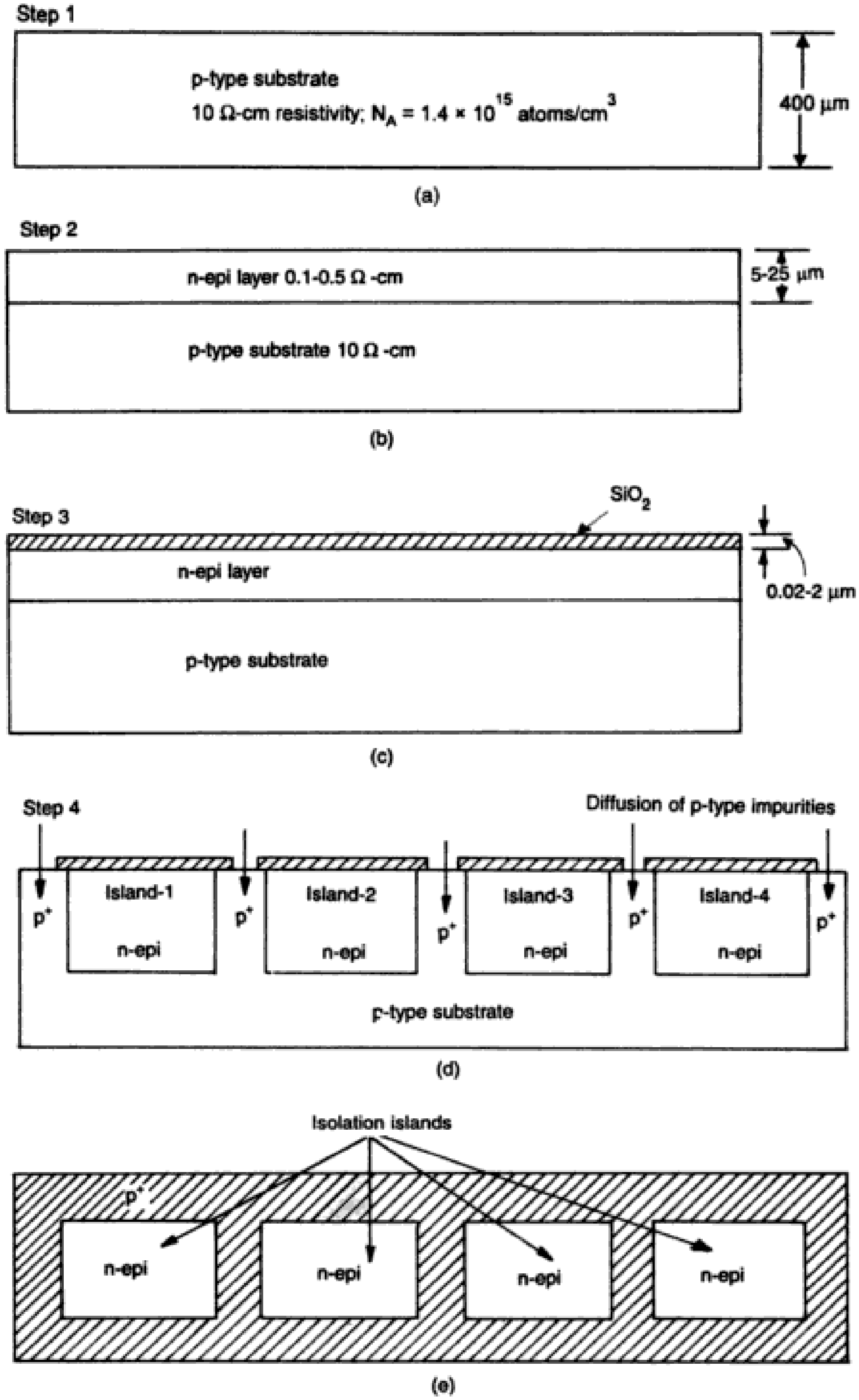


Fig. 1.14 (a-e) steps in the fabrication of the circuit shown in fig. 1.3

The concentration of acceptor atoms (N,5 x 10 cm) in the region between isolation islands is generally kept higher than p-type substrate for which N, 1.4 10 atoma/em. This ensures that the depletion region of the reverse biased diode will not extend into p region to the extent of electrically connecting the two isolation islands. There will, however, be a significant amount of barrier or transition capacitance present as a by product of the isolation diffusion. The top view of the isolation islands is depicted in Fig. 1.14 (e)

Step 5: Base Diffusion

Refer to Fig. 1.14 (0. A new layer of SiO, is grown over the entire wafer and a new pattern of openings is formed using photolithographic technique. Now, p-type impurities, such as boron, are diffused through the openings into the islands of n-type epitaxial silicon. The depth of this diffusion must be controlled so that it does not penetrate through n-layer into the substrate. This diffusion is utilized to form base region, of the transistor, resistor, the anode of the diode and junction capacitor.

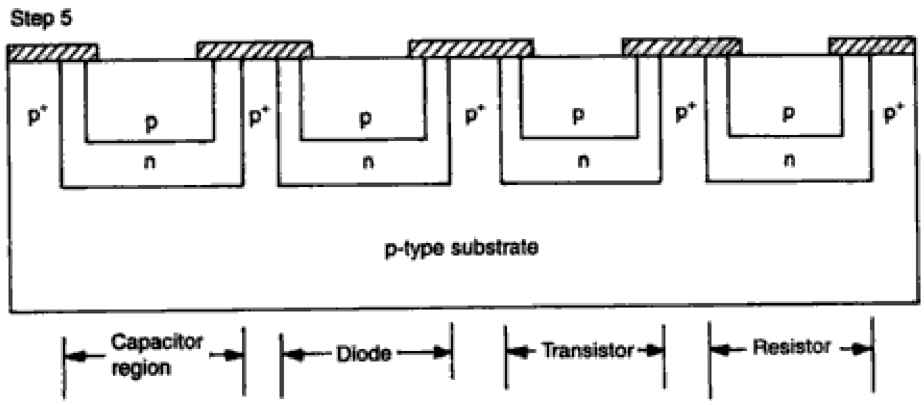


Fig 1.14 (f)

Step 6: Emitter Diffusion

Refer to Fig. 1.14 (g). A new layer of SiO, is again grown over the entire wafer and selectively etched to open a new set of windows and n-type impurity (phosphorus) is diffused through them. This forms transistor emitter and cathode region of diode. Windows (W, W, etc.) are also etched into n-region where contact is to be made to the n-type layer. Heavy concentration of phosphorus (n") is diffused into these regions simultaneously with the emitter diffusion. The reason for using heavily doped n-regions can be explained as follows:

Aluminium, normally used for making interconnections, is a p-type impurity in silicon, and can produce an unwanted rectifying contact with the lightly doped n-material. However, heavy concentration of phosphorous (-2x 10 cm) doping causes a high degree of damage to the Si-lattice at the surface, thus effectively making it semi-metallic This n layer thus makes a good ohmic contact with the Al-layer. The top view corresponding to Fig 1.14 (g) is shown in Fig. 1.14 (h).

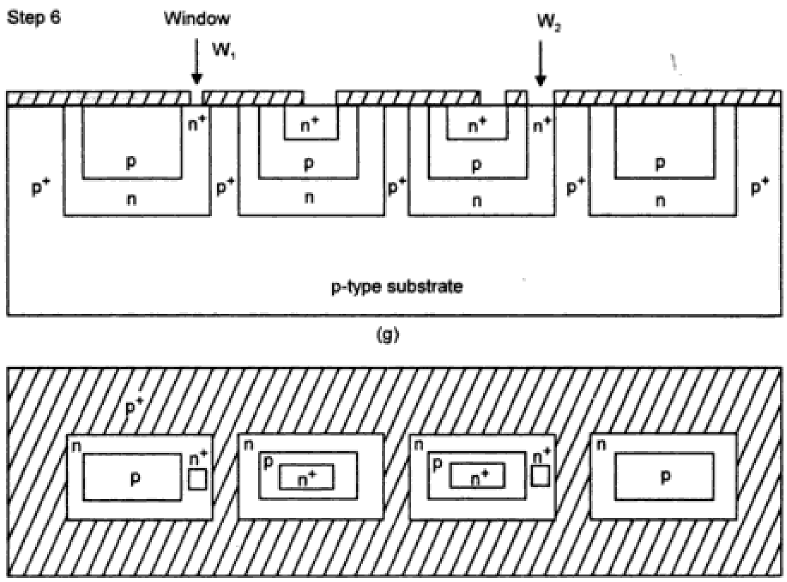


Fig 1.14 (g-h)

Step 7: Aluminium Metallization

Now the IC chip is complete with all the active and passive devices and only interconnections between the various components have to be made in accordance with the circuit in Fig. 13. The chip is further subjected to the process of the formation of a new Si0, layer and masked etching to open a new set of windows at the points where contacta have to be made. Now, a thin even coating of aluminium is vacuum-deposited over the entire surface of the wafer. The interconnection pattern between the components is then formed by photo resist techniques. The undesired aluminium areas are etched away leaving a pattern of interconnections between transistor, resistor, diode and capacitor as shown by the cross sectional and top view in Fig. 1.14 (i) and Fig. 1.14 respectively.

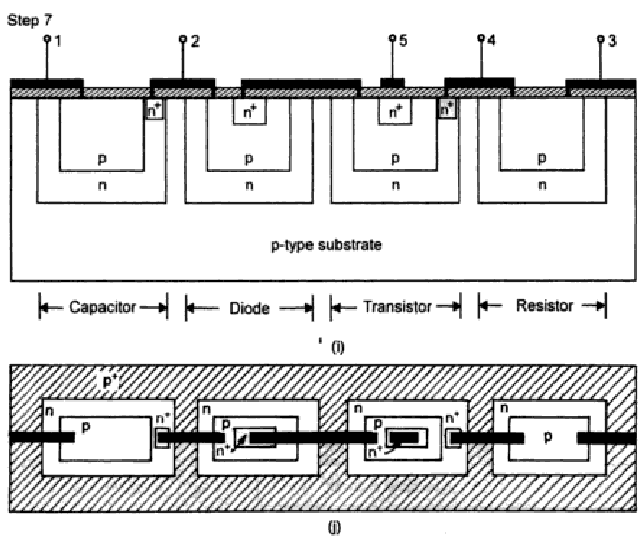
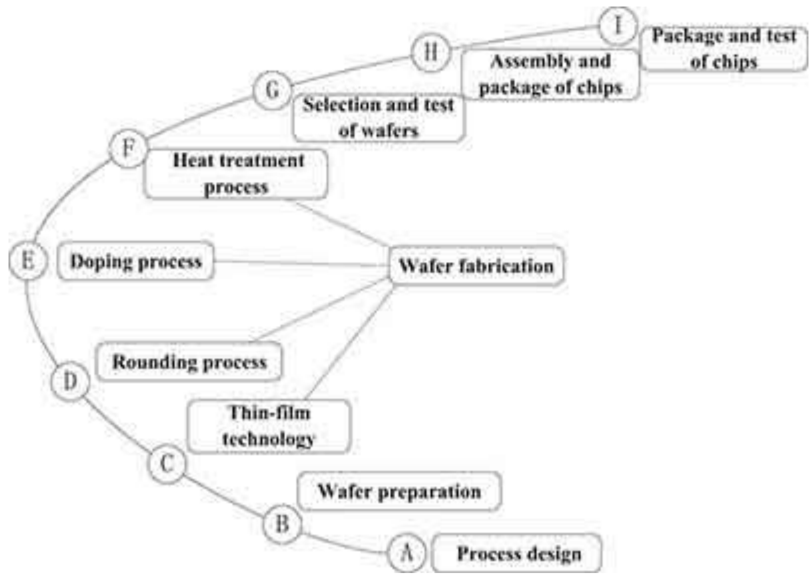


Fig 1.14 (i-j)

The Integrated Circuit Fabrication Process



The Integrated Circuit (IC) Fabrication process is a multistep manufacturing process that starts with the wafer preparation process. The silicon wafer is then cleaned and etched. After that, an oxide layer is deposited as a protective layer. A dopant is then shot at the required location to enhance the IC's performance. Packaging follows, and the finished products are ready for use. The final step of the IC fabrication process is to test the circuits to ensure their quality.

In the Integrated Circuit Fabrication process, a thin round silicon wafer is cut from the **silicon ingot** using a precision cutting machine called a wafer slicer. The silicon surface of the wafer is polished and is then fabricated into the circuit. Once the final circuit is designed, the integrated circuit is tested and shipped. The next step is to ship the chip to the customer. In most cases, the chip is sent in an antistatic plastic bag for shipping.

The process begins by cutting a thin, round **silicon wafer** from a silicon ingot using a precision cutting machine called a wafer slicer. Once the wafer is etched, it is shipped in anti-static bags. Then, the wafer is deposited on an anti-static carrier. The silicon surface is then polished to create the final circuit. It is a complex process, involving multiple steps.

The first step of the IC fabrication process is wafer preparation. The wafer material is cut, shaped, and polished. In some cases, it is necessary to modify the wafer to make it usable. The oxidation process is the next step. This involves adding oxygen to silicon. Silicon dioxide is formed from this reaction, and the silicon dioxide is removed. **Dry oxidation** is preferred because it is less expensive and the final product is more reliable.

The Integrated Circuit fabrication process starts by cutting a thin silicon ingot into individual IC chips. The silicon ingot is a piece of silicon. Then, the wafer is cut into small pieces by a precision cutting machine. These chips are then assembled and tested. Typical ICs have a variety of characteristics. For instance, a semiconductor chip can have millions of transistors. The entire fabrication process can take several months.

The ICs are made of a large number of tiny components. The ICs are fabricated in different sizes. The process is done by applying a thin layer of viscous liquid over a silicon ingot. The layer is then polished to make it easier to create individual chips. The semiconductors used in the ICs are called'silicon'. They are a material composed of a layer of **silicon**.

The semiconductor industry uses a process known as lithography to manufacture the ICs. This process begins by cutting a thin round silicon wafer. This silicon is then sliced into thin slices, which are approximately 0.004 to 0.025 inches in thickness. The layers are then assembled in a production line, where they are tested and inspected. The silicon is then cleaned and polished, and the resulting ICs are ready for implantation.

After the wafer is cleaned and polished, the process begins. The first step in this process is the creation of circuit patterns. This step involves cutting the silicon wafer into thin slices, which are 0.004 to 0.025 inches thick. The traces are then etched with a mask to make them appear as a square or rectangle. Once the pattern is formed, the wafer is placed in the etching chamber, and the light is then reflected back onto the silicon surface.

ICs are made of layers of silicon and other semiconducting materials. The layers can range from 0.000005 to 0.1 mm thick and may have as many as 30 layers. These layers are then etched with geometric shapes and lines. This process creates circuits with a multitude of functions. One of the biggest advantages of ICs is the programming capabilities they offer. Moreover, integrated circuits can be programmed to perform complex tasks.

ICs are a series of interconnected electronic circuits mounted on a tiny chip. Integrated circuits are made of various materials. The materials used in the fabrication process are carbon steel, silver, copper, magnesium, and silicon. During the IC fabrication process, they are stacked one above the other. A single semiconductor device can have as many as ten layers. However, a high-performance IC may require more than one material

References:

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